# Livia Popa

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## Education

**Cornell University** — Ithaca, NY

*Doctor of Philosophy (PhD) in Statistics* — *Aug. 2022 – May 2027*

- Advisors: Sumanta Basu and Martin Wells
- GPA: 3.953/4.00
- Activities: President of Cornell Statistics Graduate Society (SGS), Social Chair SGS
- First Year Support: Cornell University Graduate Fellowship (2022-2023)

**North Carolina State University** — Raleigh, NC

*Bachelor of Science (BS) in Statistics and Economics (Double Major), Minor in mathematics* — *Aug. 2019 – May 2022*

- GPA: 3.97/4.00, *summa cum laude*
- Activities: Treasurer of Club Tennis at NC State

## Research Experience

**Graduate Researcher** — May 2023 – Present

*Cornell University Bowers CIS* — *Ithaca, NY*

*Time Series and Machine Learning Project* — *Jan 2025 – Present*

- Developing general machine learning frameworks to estimate solutions to integro-differential equations (IDEs), leveraging reservoir computing methods such as Echo State Networks and Nonlinear Vector AutoRegression (NVAR), with applications in financial modeling and biological systems.
- Designing and implementing methods to quantify uncertainty in reservoir computing models, including conformal prediction and Bayesian regularization techniques, to provide robust forecast intervals and reliability estimates.
- Extending uncertainty quantification approaches to NVAR (also known as NGRC), enabling interpretable and probabilistically grounded confidence bounds for time series forecasts in high-noise environments.

*Bayesian Time Series Project* — *May 2023 – March 2025*

- Conducted Bayesian time series and high-dimensional regression research with applications to Omics, Diabetes, and Econometrics data, focused on identifying biomarkers from short-term longitudinal studies.
- Developed and implemented a univariate Bayesian regression approach using Zellner's $g$-prior with data-adaptive choices of $g$ (e.g., $g = n$, SURE-optimal $g$) and a multivariate Bayesian Group LASSO with spike-and-slab priors to improve inference, perform group selection, and quantify uncertainty.
- Achieved **<10% false positive rates** and high sensitivity using Bayes Factors for variable inclusion, outperforming standard mixed-effects models in biomarker selection and interpretability.
- Created biomarker network graphs using Laplacian-penalized priors, aiding in biological interpretation and visualizing metabolite relationships.

**Open Climate Data Science Undergraduate Research Assistant** — Sep. 2021 – May 2022

*North Carolina State University* — *Raleigh, NC*

- Designed and led hands-on workshop tutorials in **RStudio** to teach the fundamentals of Random Forest models, including bagging, variance reduction, and feature importance.
- Developed interactive scripts and walkthroughs using `randomForest` and `caret` packages to guide participants through model training, tuning, and evaluation.
- Analyzed post-workshop survey data using `tidyverse` tools; summarized key insights in a final report to assess workshop effectiveness and support future improvements.
- Taught Random Forest modeling to **50+ participants**, leading to a **40% increase** in self-reported confidence using ML techniques in R, as measured via pre- and post-surveys.

**Undergraduate Researcher** — Aug. 2020 – May 2022

*Functional Data Analysis Lab Group at North Carolina State University* — *Raleigh, NC*

- Completed hands-on machine learning projects using Python, with a focus on scikit-learn and TensorFlow libraries, including supervised and unsupervised learning workflows.
- Contributed to research on Functional Graphical Models and Principal Component Analysis (PCA) for high-frequency time series data, evaluating statistical dependencies and dimensionality reduction in functional settings.
- Collaborated with faculty and graduate students on methodological development and literature review related to functional time series, covariance estimation, and graphical model selection.

## Industry Experience

**Quantitative Research Intern** May 2026 – Aug. 2026
*Quantbot Technologies, LP* *New York, NY*

- Evaluated ESN, NGRC, Ridge, OLS, and XGBoost on approximately **24M** out-of-sample stock-horizon forecasts across **500** liquid equities and five horizons (10 min–EOD), using a full-year 2013 holdout.
- Reproduced an **8–13**% MSFE improvement of ESN over OLS at intraday horizons and showed that regularized Ridge recovered nearly all of the gain, matching ESN within **0.5–3.5**% and identifying linear memory as the primary source of predictive performance.
- Developed forecast- and feature-blending studies spanning **13** futures roots, **8** input forecasts, four horizons, and **9** model configurations; built hyperparameter-search infrastructure with CMA-ES/Optuna, **48**-configuration NGRC grid searches, shared caching, checkpointing, and memory controls.

**Data Science Institute Graduate Intern** May 2025 – Aug. 2025
*Lawrence Livermore National Labs* *Livermore, CA*

- Applied supervised ML and Bayesian statistical techniques—including Gradient boosting, Random Forests, and Support Vector Machines (SVM)—to classify uranium ore concentrate (UOC) samples.
- Improved classification accuracy on the dataset of interest from **57% to 85%** using Random Forests, and from **80% to 97%** using Gradient Boosting, significantly outperforming baseline models.
- Quantified uncertainty in classification decisions using ensemble learning and probabilistic techniques to support risk-aware nuclear material source attribution.
- Built a custom Partial Least Squares Discriminant Analysis (PLS-DA) classifier using Python's `scikit-learn` `PLSRegression` class to analyze and classify high-dimensional nuclear forensic data.
- **Computer Vision Project – Modal-to-Amodal Masking in Video:** Collaborated with a multidisciplinary team on a task focused on modal-to-amodal masking for images and videos, improving occlusion reasoning in object recognition systems.

**Shelton Jones Statistical Intern** June 2021 – Aug. 2021
*RTI International* *Research Triangle Area, NC*

- Developed robust SAS programs to clean, validate, and analyze patient survey data from federally funded health centers, enabling accurate insights into patient demographics, satisfaction, and health outcomes.
- Used R to extract and transform data from the U.S. Census Bureau's public datasets, integrating key demographic and socioeconomic indicators into the project's analytical pipeline.
- Built customized geography tables to improve the quality of auxiliary information appended to the sampling frame, enhancing coverage estimates and improving downstream statistical modeling.

## Teaching/ Administrative Experience

**Lead Teaching Fellow** Jan. 2025 – May 2025
*National Education Equity Lab*

- Lead Weekly Discussion sections and hold office hours for high school students who are enrolled in Cornell Essential Statistics and Data Science Course with a focus on R programming skills and introductory statistics knowledge

**Graduate Project Specialist** Jan. 2024 – Dec. 2024
*Cornell Bowers CIS Office of Diversity, Equity, Inclusion and Belonging (DEIB) University* *Ithaca, NY*

- Contributed to the research, data collection, analysis, and reporting for Cornell Bowers CIS DEI's programs, events, and services
- Supported the Assistant Director for DEI in developing, facilitating, and evaluating spring and fall programming. Collaborated with Cornell Bowers CIS leaders, partners, and student organizations

**Head Teaching Assistant** Multiple Dates
*Cornell University* *Ithaca, NY*

- STSCI Probability Models and Inference *Spring 2026*
- STSCI Introductory Statistics for Biology, *Spring 2024, Fall 2026*
- ILRST Introductory Statistics,*Fall 2023, Winter 2024, Summer 2024, Winter 2025*
- PSYCH 5750 Quantitative Methods I, Grader, *Fall 2023*

**Data and Office Assistant** Jan. 2020 – March 2021
*NCSU Office of Undergraduate Research* *Raleigh, NC*

- Filtered and cleaned Qualtrics data from Research Student Surveys to help give insights for future changes
- Developed a new database for storing undergraduate student information

## Honors and Awards

Artificial Intelligence for Sustainability (AI4S) NRT Fellowship Spring 2027

G-Research Women in Quant Finance 2026 SRPreview program

G-Research Spring into Quant Finance 2026 Doctoral Fellowship

Schonfeld 2025 Early Engagement (SEE) Summit – PhD Datathon Participant

Distinguished in Leadership and Service Award, Cornell Bowers CIS Office of Diversity Equity and Inclusion

Cornell Graduate School Fellowship

North Carolina State University, College of Sciences Dean's List

North Carolina State University, Poole College of Management Dean's List

## Publications and Preprints, Conference Presentations

**Popa, L.**, Basu, S., & Wells, M. T. (2026). *Uncertainty Quantification of Next Generation Reservoir Computing with Applications to Memory-Driven Dynamical Systems.* arXiv:2609.32169. *Submitted to ICLR 2027.*

**Popa, L.**, Basu, S., Lee MH., & Wells, M. *Uncertainty Quantification for Next Generation Reservoir Computing with Applications to Memory Driven Systems.* Joint Statistical Meetings. Boston, MA. August 2026. *Paper Presentation*

**Popa, L.**, Basu, S., Lee MH., & Wells, M. (2025). *BLOG: Bayesian Longitudinal Omics with Group Constraints.* 10.1515/sagmb-2025-0039. *Statistical Applications in Genetics and Molecular Biology.*

**Popa L.**, Caseres J. *Machine Learning Classifiers for Nuclear Forensics: Applications to Uranium Ore Concentrate Datasets* Cornell University Day of Statistics and Data Science. Ithaca, NY. Sep. 2025. *Poster Presentation*

**Popa, L.**, Basu, S., Lee MH., & Wells, M. *BLOG: Bayesian Longitudinal Omics with Group Constraints.* Joint Statistical Meetings. Nashville, TN. August 2025. *Paper Presentation*

**Popa L.**, Caseres J. *Machine Learning Classifiers for Nuclear Forensics: Applications to Uranium Ore Concentrate Datasets* Lawrence Livermore National Labs Summer Student Showcase. Livermore, CA. August 2025. *Poster Presentation*

**Popa L.**, Basu S., Lee MH., Wells M. *High- Dimensional Bayesian Regression for Longitudinal Omics Data and Reservoir Computing.* North Carolina State University Functional Data Analysis Seminar Series. Raleigh, NC. March. 2025. *Invited Seminar Talk*

**Popa L.**, Basu S., Lee MH., Wells M. *High- Dimensional Bayesian Regression for Short - Term Longitudinal Data.* Cornell University Day of Statistics and Data Science. Ithaca, NY. Sep. 2024. *Poster Presentation*

**Popa L.**, Basu S., Lee MH., Wells M. *Bayesian Regression for Short - Term Longitudinal Data.* Joint Statistical Meetings. Portland, OR. August 2024. *Poster Presentation*

## Coursework

**Statistics**: Numerical Optimization and Statistical Methods in Data Science, Theory of Linear Models, Distributions Convergence and Estimation Theory, Nonparametric Estimation and Minimax Theory, Causal Inference and Machine Learning, Statistical Consulting I, Statistical Consulting II
**Mathematics**: Real Analysis, Measure Theoretic Probability, Differential Equations, Linear Algebra, Statistical Probability
**Computer Science**: Matrix Computations, Understanding Machine Learning
**Economics:** Econometrics II, International Finance, Long-term Actuarial Models, Labor Economics, Public Finance, Macroeconomics, Microeconomics

## Technical Skills

**Languages**: Python, R, SAS, SQL, HTML/CSS, JMP
**Data Science/Machine Learning**: Pandas, NumPy, Matplotlib, PyTorch, SciPy, scikit-learn, TensorFlow, WordPress, Jupyter Notebook
**Developer Tools**: Git, VS Code, Visual Studio, PyCharm, Latex